Working with fractions

QUICK REFERENCE | Suggested grades 3-6 | Math

Use equal-sized parts. The denominator tells the size of the parts.

Name the parts

In 3/4, the numerator 3 counts the parts.

The denominator 4 means the whole is split into 4 equal parts.

Make an equivalent fraction

Multiply or divide the top and bottom by the same nonzero number.

1/2 = 2/4 = 3/6. Divide 6/8 by 2 on top and bottom to get 3/4.

Add or subtract: use a common denominator

1/2 + 1/3 = 3/6 + 2/6 = 5/6.

3/4 - 1/2 = 3/4 - 2/4 = 1/4.

Add or subtract the numerators. Keep the common denominator.

Multiply across

2/3 x 3/4 = (2 x 3)/(3 x 4) = 6/12 = 1/2.

Divide: multiply by the reciprocal

3/4 divided by 1/2 = 3/4 x 2/1 = 6/4 = 1 1/2.

Do not divide by zero.

Check the whole

Compare fractions of the same-sized whole. Draw fraction bars if you need a picture.

OPP